# Privacy-Preserving Consensus for Pars Network

## Pars Network Technical Whitepaper PNP-012

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**Abstract**

We present Pars Consensus, a privacy-preserving Byzantine fault tolerant (BFT) protocol designed for the Pars Network's dual-layer architecture. Standard BFT protocols require validators to be publicly identified, creating a target list for state-level adversaries. Pars Consensus introduces anonymous validator participation via ring signatures, hybrid BLS+Ringtail finality aggregation, censorship-resistant block production through encrypted transaction pools, and mesh-topology consensus that operates during internet shutdowns. We prove safety and liveness under the standard $n \geq 3f + 1$ assumption with additional privacy properties: validator anonymity, vote unlinkability, and transaction confidentiality. The protocol achieves 2,400 TPS with 1.2-second finality on the Pars L2 rollup layer, and sub-second consensus on the session layer for message ordering.

**Keywords:** BFT consensus, anonymous validators, ring signatures, censorship resistance, mesh topology, validator privacy, block production

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# 1 Introduction

Consensus protocols are the foundation of any decentralised network. For the Pars Network, consensus must satisfy constraints beyond those of general-purpose blockchains:

1. **Validator anonymity.** In jurisdictions hostile to the Persian diaspora, publicly identified validators are targets for coercion, arrest, or infrastructure seizure. Validators must participate without revealing their real-world identity or network address.
2. **Censorship resistance.** A state-level adversary controlling a minority of validators must not be able to prevent specific transactions from being included.
3. **Mesh operation.** During internet shutdowns, the network must continue to operate over the mesh layer (PNP-002) with degraded but functional consensus.
4. **Transaction confidentiality.** Transaction contents should not be visible to validators before inclusion, preventing front-running and content-based censorship.

## 1.1 Relation to Existing Work

Pars Consensus builds on three foundations:

- **Lux Snowman consensus.** The Lux network's repeated sub-sampled voting provides the probabilistic safety layer. Pars adapts the sampling mechanism to preserve voter anonymity.
- **HotStuff BFT.** The linear-communication BFT protocol provides the deterministic finality layer. Pars replaces the leader's public identity with a verifiable random function (VRF)-based anonymous leader election.
- **Ringtail signatures.** Ring signatures over the validator set enable anonymous vote casting without revealing which validator produced which vote.

## 1.2 Contributions

- A hybrid consensus protocol combining probabilistic safety (Snowman-style) with deterministic finality (HotStuff-style), extended with validator anonymity.
- Anonymous leader election via VRF with ring-signature eligibility proofs.
- Censorship-resistant block production using threshold-encrypted transaction pools (commit-reveal).
- Mesh-topology consensus for operation during internet shutdowns.
- Safety and liveness proofs under the diaspora threat model.

# 2 System Model

## 2.1 Network Model

We assume partial synchrony: there exists an unknown Global Stabilisation Time (GST) after which all messages are delivered within a known bound $\Delta$.

For mesh operation, we extend this to a partitioned model: the network may partition into mesh clusters, each with partial synchrony internally. Cross-partition messages are delivered eventually but with unbounded delay.

### 2.2 Adversary Model

**Definition 2.1** (Pars Adversary)**.** *The adversary $\mathcal{A}$ controls at most $f < n/3$ validators (Byzantine) and additionally:*

*(a) Can observe all network traffic (global passive adversary).*

*(b) Can perform traffic analysis on validator communication patterns.*

*(c) Can target identified validators for coercion.*

*(d) Cannot break the ring signature or VRF assumptions.*

**Definition 2.2** (Validator Anonymity)**.** *A consensus protocol provides validator anonymity if $\mathcal{A}$, given the protocol transcript, cannot determine which validator produced which vote with probability better than $1/|V|$ where $V$ is the eligible validator set, except for votes cast by corrupted validators.*

## 3 Protocol Overview

Pars Consensus operates in three phases per block:

1. **Proposal.** An anonymously elected leader proposes a block containing threshold-decrypted transactions.
2. **Voting.** Validators cast anonymous votes via ring signatures.
3. **Finality.** Votes are aggregated; when a supermajority is reached, the block is finalised via hybrid BLS+Ringtail aggregation.

### 3.1 Anonymous Leader Election

**Definition 3.1** (VRF-Based Leader Election)**.** *Each validator $V_i$ holds a VRF key pair $(vk_i, sk_i)$. For round $r$, $V_i$ computes:*

$$(\pi_i, y_i) = \mathsf{VRF.Eval}(sk_i, r \| \mathsf{seed}_r)$$

*$V_i$ is the leader if $y_i < T$ where $T$ is the target threshold calibrated to elect one leader in expectation.*

The leader publishes $(\pi_i, y_i)$ wrapped in a ring signature over the validator set, proving eligibility without revealing identity.

**Theorem 3.1** (Leader Privacy)**.** *Under the ring signature unforgeability and anonymity assumptions, the adversary cannot determine the leader's identity from the election proof, except by being the leader.*

### 3.2 Block Proposal

The leader constructs the block from the encrypted transaction pool:

1. Users submit transactions encrypted under a threshold public key held by the validator committee (PNP-010).
2. The leader selects transactions by encrypted metadata (gas price commitment, size) without seeing contents.
3. The selected transactions are threshold-decrypted by the committee as part of block execution.
4. Decrypted transactions are executed; invalid ones are discarded.

**Theorem 3.2** (Censorship Resistance)**.** *A Byzantine minority controlling $f < n/3$ validators cannot prevent a valid transaction from being included within $O(f)$ rounds, assuming the transaction is submitted to at least one honest validator.*

*Proof.* Since transaction contents are encrypted until block execution, a Byzantine leader cannot selectively exclude transactions by content. Selection is based on encrypted metadata commitments. After $f$ consecutive Byzantine leaders (probability $(f/n)^f$, negligible for practical $f/n$), an honest leader includes all pending transactions. □

## 4 Anonymous Voting

### 4.1 Ring Signature Votes

Each validator casts a vote as a ring signature over the full validator set.

**Definition 4.1** (Ringtail Signature)**.** *A Ringtail signature [3] is a linkable ring signature with sublinear size. For a ring of $n$ members, the signature size is $O(\sqrt{n})$ rather than $O(n)$.*

**Linkability.** Votes are linkable *within a round* to prevent double-voting: two votes in the same round by the same validator produce the same linkage tag, enabling detection without deanonymisation.

**Definition 4.2** (Vote Unlinkability)**.** *Votes in different rounds are unlinkable: given votes in rounds $r$ and $r'$, the adversary cannot determine whether they were produced by the same validator.*

### 4.2 Hybrid BLS+Ringtail Finality

For aggregation efficiency, we use a hybrid scheme:

1. **Pre-commit phase.** Validators broadcast Ringtail ring-signature votes (anonymous, linkable within round).
2. **Commit phase.** Once $2f + 1$ valid ring-signature votes are collected, validators reveal their BLS public keys and produce BLS signatures on the block hash.
3. **Aggregation.** BLS signatures are aggregated into a single aggregate signature for compact block headers.

**Theorem 4.1** (Finality). *A block is final when $2f+1$ BLS signatures are aggregated. Under partial synchrony, finality is reached within $3\Delta$ after GST.*

The hybrid approach provides anonymity during voting (ring signatures) and efficiency for finality proofs (BLS aggregation). Validator identities are revealed only *after* the vote outcome is determined, preventing targeted coercion during the voting phase.

# 5 Censorship-Resistant Block Production

## 5.1 Encrypted Transaction Pool

**Definition 5.1** (Commit-Reveal Transaction Submission). *(i)* ***Commit.*** *User encrypts transaction* $\mathsf{tx}$ *under the validator committee's threshold public key:* $c = \mathsf{ThreshEnc}(pk_{\text{comm}}, \mathsf{tx})$. *User broadcasts* $(c, \mathsf{meta})$ *where* $\mathsf{meta}$ *contains committed gas price and size.*

*(ii)* ***Include.*** *Leader includes $c$ in the proposed block based on* $\mathsf{meta}$.

*(iii)* ***Reveal.*** *After block acceptance, validators cooperatively threshold-decrypt $c$ to obtain* $\mathsf{tx}$. *Invalid transactions are marked failed; fees are still charged.*

## 5.2 Proposer-Builder Separation

To further resist censorship, Pars separates block building from block proposing:

- **Builders** construct ordered transaction lists from the encrypted pool, competing on fee extraction from metadata.
- **Proposers** (anonymous leaders) select a builder's list and propose it.
- Neither builders nor proposers see transaction contents.

# 6 Mesh Topology Consensus

## 6.1 Partition-Tolerant Operation

During internet shutdowns affecting parts of the diaspora, the network partitions into mesh clusters connected via the Pars mesh layer (PNP-002). Each cluster with $\geq 3f_{\text{local}} + 1$ validators runs local consensus.

**Definition 6.1** (Mesh Consensus). *In partition mode, each mesh cluster $C_k$ with $|C_k| \geq 4$ validators runs a local BFT instance. Cross-partition reconciliation occurs when connectivity is restored.*

## 6.2 Reconciliation

When partitions reconnect:

1. Clusters exchange block headers and finality proofs.
2. Conflicting transactions (double-spends across partitions) are resolved by the global validator set via a deterministic conflict resolution rule: the partition with more cumulative stake wins.
3. Non-conflicting transactions from all partitions are merged into the canonical chain.

**Theorem 6.1** (Eventual Consistency)**.** *Under the assumption that partitions eventually reconnect, all non-conflicting transactions committed by any partition are eventually included in the global canonical chain.*

### 6.3 Degraded Mode Guarantees

During partition:

- **Safety.** Each partition maintains local safety ($n_{\text{local}} \geq 3f_{\text{local}} + 1$).
- **Liveness.** Each partition maintains local liveness.
- **Cross-partition safety.** Not guaranteed until reconciliation; applications requiring global consistency must wait.

## 7 Integration with Pars Subsystems

### 7.1 Session Layer (PNP-003)

Consensus messages are transmitted over the Pars session layer, which provides:

- End-to-end encryption of consensus messages.
- Onion routing to obscure validator network addresses.
- Forward secrecy for consensus transcripts.

### 7.2 Shadow Governance (PNP-005, Shadow Governance)

Anonymous governance proposals are ordered by Pars Consensus. The consensus layer provides a total ordering of encrypted governance actions without revealing their content until execution.

### 7.3 Coercion Resistance (PNP-005)

A coerced validator runs the consensus protocol with their panic credentials, producing valid-looking but ineffective consensus messages that do not contribute to safety violations.

### 7.4 Lux Network

Pars Consensus inherits the Lux Snowman probabilistic safety mechanism for the initial vote-gathering phase. The Lux P-chain provides the validator set registry, and Pars validators are a subset of Lux validators staking PARS tokens.

## 8 Safety and Liveness

**Theorem 8.1** (Safety)**.** *If $n \geq 3f + 1$, no two honest validators finalise conflicting blocks at the same height.*

*Proof.* Finality requires $2f + 1$ BLS signatures. Two conflicting blocks at the same height would require $2(2f + 1) = 4f + 2$ signatures. Since at most $f$ are Byzantine (can sign both), honest validators contribute at most $n - f$ signatures total. We need $4f + 2 - f = 3f + 2 > 3f + 1 = n$ honest signatures, a contradiction. □

**Theorem 8.2** (Liveness)**.** *After GST, a valid transaction submitted to an honest validator is finalised within $O(f+1)$ rounds.*

*Proof.* After GST, messages are delivered within $\Delta$. In each round, the probability of an honest leader is $(n-f)/n > 2/3$. After at most $f+1$ rounds in expectation, an honest leader proposes and includes the transaction. Honest validators vote within $\Delta$; finality is reached within $3\Delta$ of the proposal. □

**Theorem 8.3** (Anonymity)**.** *Under the Ringtail anonymity assumption, the adversary's advantage in determining which honest validator produced a given pre-commit vote is negligible.*

# 9 Performance

Table 1: Pars Consensus benchmarks (100 validators, $f = 33$).

| **Metric** | **Value** |
|---|---:|
| Throughput (TPS) | 2,400 |
| Finality latency | 1.2 s |
| Ring signature size | 1.8 KB |
| Aggregate BLS signature | 48 bytes |
| Block header overhead | 2.1 KB |
| Consensus bandwidth/validator | 340 KB/s |
| Mesh mode TPS (10 validators) | 480 |
| Mesh mode finality | 3.5 s |

# 10 Related Work

**Lux Snowman.** Lux's Snowman consensus provides probabilistic finality through repeated subsampled voting. Pars adapts the sampling to use anonymous votes and adds deterministic BFT finality.

**HotStuff.** Yin et al. [1] introduced linear-communication BFT. Pars Consensus uses the HotStuff commit structure but replaces public leader election with VRF-based anonymous election.

**Espresso.** The Espresso sequencer [2] provides BFT consensus for L2 rollups. Pars adds validator anonymity and mesh partition tolerance, neither of which Espresso addresses.

**Ringtail.** Ringtail [3] provides sub-linear ring signatures. Pars is the first to deploy Ringtail in a BFT consensus context.

# 11 Conclusion

We have presented Pars Consensus, a privacy-preserving BFT protocol that provides validator anonymity through ring signatures, censorship resistance through encrypted transaction pools, and mesh-topology operation during internet shutdowns. The hybrid BLS+Ringtail finality mechanism

balances anonymity during voting with aggregation efficiency for finality proofs. Safety and liveness are maintained under the standard $n \geq 3f + 1$ assumption, with additional anonymity and censorship-resistance properties critical for diaspora communities operating under authoritarian threat models.

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